Simone Rea

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Education

Eindhoven University of Technology – TU/e — **Eindhoven, Netherlands**
Incoming student accepted in Erasmus Program — February 2027 – July 2027

Yonsei University — **Seoul, Korea**
Incoming student accepted in Overseas Program — September 2026 – December 2026

Alma Mater Studiorum – University of Bologna — **Bologna, Italy**
Bachelor's Degree in Computer Engineering — 2024 – Present

- Current GPA: 29.35/30 (equivalent to **3.926/4**)

Istituto Tecnico-Scientifico Ettore Majorana — **Orvieto, Italy**
Traditional Scientific High School Diploma — 2019 – 2024

- Graduated with **100/100 cum laude**

Experience

Biozentrum – University of Basel — **Basel, Switzerland**
Summer Researcher, Engel's Lab — July 2026 – September 2026

- Performed **expansion microscopy (ExM)** on diverse biological samples (*Chlorella*, *Chlamydomonas*, *Auxenochlorella*, plant leaf tissue) and characterized subcellular structures via **confocal/widefield fluorescence imaging** with multiple molecular markers.
- Investigated a recurring **"Fusilli Fold" structural motif** in protein sequences via a bioinformatics pipeline combining **MEME** motif discovery, **MUSCLE** sequence alignment, **HMM** profile construction, and **Jackhmmer** database searches across diverse organisms.
- Implemented and adapted **CryoTIGER**, a deep-learning-based tilt-interpolation tool for cryo-electron tomography, to improve 3D reconstruction quality on tomographic datasets

ADM Team – Alma Mater Studiorum — **Bologna, Italy**
Software Engineering Member — November 2024 – November 2025

- Developed backend services in **Go** to implement a DNS system backed by **PowerDNS**.
- Implemented and tested HTTP APIs using `curl` for request validation and debugging.
- Acquired practical knowledge of distributed **system architecture** and service-oriented design.
- Set up **Prometheus** for monitoring, metrics collection, and basic alerting.
- Worked on application **deployment** and environment configuration in Linux-based systems.

Unibo Motorsport — **Bologna, Italy**
Novice Member – Driverless Department — March 2025 – June 2025

- Collaborated in a multidisciplinary university team.
- Contributed to the Software Infrastructure sub-department by utilizing **C#** and **Unity** to support the development of a driving simulator.
- Designed and implemented a custom control interface for an autonomous robot, leveraging **Python** and **ROS2** to meet specific sponsor requirements.

Projects

- CyberChallenge program – selected among the **top 25 Computer Science students** at the University of Bologna for this highly competitive national cybersecurity training program
- High School Math Olympics

Certifications

- IELTS Academic – Overall Band Score: 7.5

Technical Skills

Programming Languages: Python, Rust, Java, C++, C, JavaScript, Go, SQL
Web Technologies: HTML, CSS, Node.js
Tools & Software: Unity, Figma, Photoshop, Prometheus
Systems & Concepts: Linux, ROS2, Algorithms, Operating Systems, Computer Networks, POSIX, Proxmox, LDAP